LINE FOLLOWER

Aim:

To design a completely autonomous bot capable of following a black marked path on white surface.

TEAM SPECIFICATION:

- All the participants must be from IIEST Shibpur only.
- A team should consist of 3 to 4 members only.
- Each participant is allowed to be part of only one team.
- Only one team member will be permitted inside the arena during the run.

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BOT ALLOWANCES :

- The robot must be wireless and completely autonomous, with no external control during the run.
- The robot may be circular or rectangular in shape.
- The robot must fit within the maximum dimensions of 30 cm × 25 cm × 20 cm (Length × Width × Height) at all times during the run.
- Maximum weight of the bot must not exceed 2kg.
- Ready-made robots and drones are strictly prohibited from participating in the event.
- The robot must be primarily designed and constructed by the participating team.
- The electrical voltage present anywhere in the robot must not exceed 16.8 V DC at any point of time.
- External power sources are strictly prohibited during the competition run.
- A 220V , 50Hz AC power supply will be provided only for pre-run setup and testing; during the actual run, the robot must operate exclusively on onboard power.

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Rounds:

The competition will be conducted as 2 round competition. The exact arena for both the rounds will be revealed on the spot. It is necessary for all the participants to run their bot for both the rounds.

Start and Restarts :

1.The robot will be placed at the starting point with the referee's consent.

2.The Race Clock:

Starts when the robot crosses the start line.

Stops when the robot crosses the Finish line.

Restart Conditions :

- A restart may be requested if the robot fails to follow the line, stops during the run, or loses the line or direction.
- In the event of a restart, the robot must be placed back at the starting point.
- The Race Clock will be reset to zero for each restart.
- The Competition Clock (maximum allotted time) will continue to run and will not be reset.
- Each team is allowed only one restart during the entire competition.

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Mandatory Restart :

A restart will be enforced if:

- The robot is touched by a human without referee consent
- The robot moves out of the arena.
- The referee orders a restart.

Programming and Modifications :

- Reprogramming during the run is strictly prohibited.
- Adding or removing parts during the run is not allowed.
- Sensor adjustment is allowed only with referee approval.

Judging Criteria :

- The score of the checkpoints covered will be considered as priority.
- The time bonus will be awarded to the team based on the time taken to complete the circuit.
- The aggregate score of both the rounds will be added and then the winner will be decided based on the leaderboard that will be prepared after both the rounds.
- Penalties may be imposed in cases of manual intervention, violation of competition rules, or deviation of the robot from the track.
- The decisions of the judges and referees shall be final and binding in all matters related to evaluation and scoring.

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- **Penalty Rules :**
- A robot will be considered to be correctly following the track only when the major portion of the line is covered by the projected area of the robot.
- a. Failure to meet this condition will attract a penalty of 25 points.
- Any physical contact (hand touch) with the robot during the run will result in a penalty of 30 points per instance.
- In the event of a technical failure during the run:
- a. A team may take a maximum of two timeouts, each with a duration of up to two minutes.
- b. There is no penalty for the first timeout but the 2nd timeout will incur a penalty of 150 points.
- Timeouts and restart are treated independently, and penalties for each will be applied separately.
- Teams will be allowed a after a failure.
- a. Failure to report back to disqualification.
- Teams may choose to skip stages, subject to the following penalties:
- a. First skip: 100 points
- b. Second skip: 150 points
- c. Third skip: 200 points
- d.A maximum of three skips is permitted.

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- **General Rules :**
- The robot must operate in a fully autonomous mode at all times, with no external control during the run.
- A scoreboard will be maintained throughout the competition to record the performance of all participating teams.
- The final results and winners will be declared on the day of the event based on the scores recorded on the scoreboard.
- Teams are advised not to disassemble or modify their robots until the results are officially announced.
- In case of any discrepancy or dispute, the decision of the organising committee and referees shall be final and binding.
- For a bot to be following the line, the major portion of the area covered by the line should be overlapped by the area projection of the bot 25 points will be penalized if violated.
- Every team will be provided with 220V , 50Hz stable AC supply. If you are using an external supply, the connecting wire must be slack at all point of time. A penalty of 500 points will be imposed if violated. In case of disruption due to wire enlargement, you are allowed to free your bot using your hands without changing its position and no penalty will be imposed in this case, unless any unfair means is adapted.

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- In case of technical failure during the run, you can take timeout/restart. You can take timeout of total 2 minutes to try and fix the issues. 2 timeouts per team are allowed. You can also take a restart. Only 1 restart is allowed per team. You will get 30 minutes to fix your bot, after which the bot must appear on the track. Each timeout will cost the team 150 points and the restart will also reset the point collector to 0 (no starting bonus will be there).
- You can skip a stage but you will not be awarded any points for that.
- Scores will be made public and winners will be declared publicly on the day of the event. You are advised not to disassemble the bot before the results are declared.

In case of any discrepancy the Decision of Team ROBODARSHAN will be final and binding

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